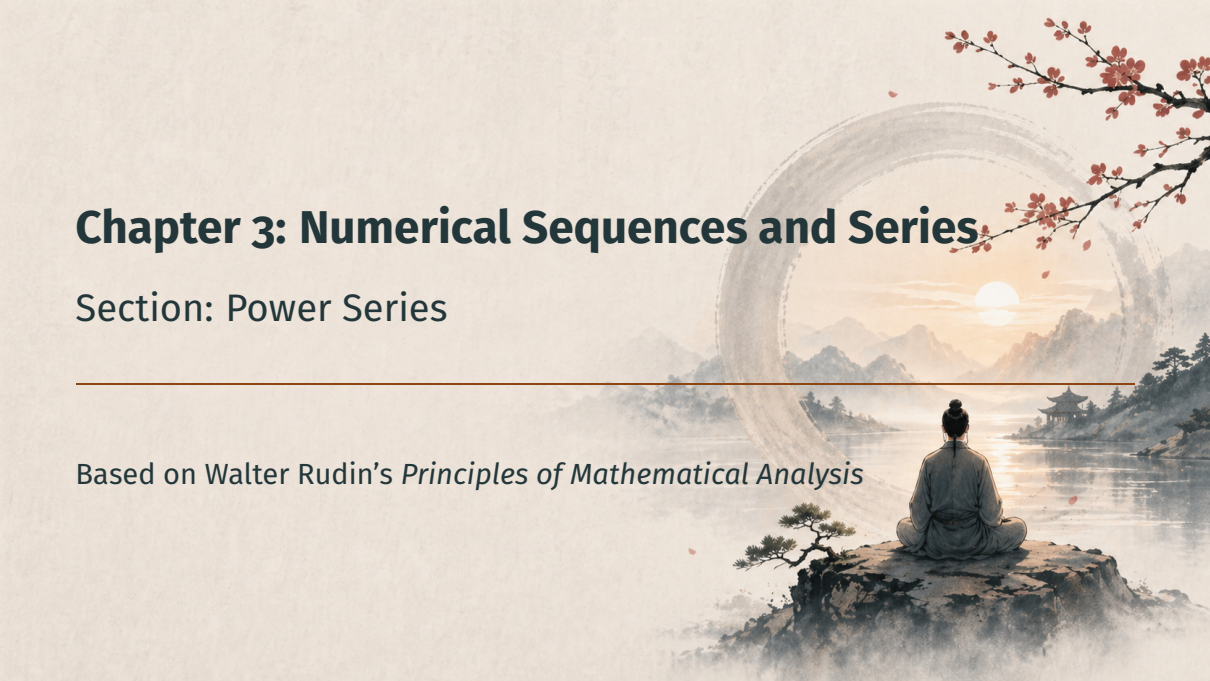


# Chapter 3: Numerical Sequences and Series

The background is a traditional Chinese ink wash painting. A person in a grey robe is seen from behind, sitting in a meditative pose on a dark, craggy rock in the foreground. They are looking out over a calm lake that reflects the sky. In the distance, misty mountains are visible. A large, faint circular frame, resembling a ring or a portal, is centered in the background. A small pavilion with a traditional roof is situated on the right side of the lake. Red cherry blossom branches with small flowers hang down from the top right corner. The overall atmosphere is serene and contemplative.

## Section: Power Series

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Based on Walter Rudin's *Principles of Mathematical Analysis*

# Outline

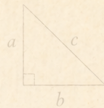
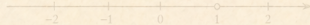
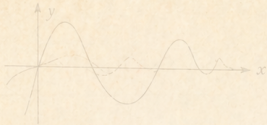
Motivation

Definition

The Radius of Convergence

Examples

Summary



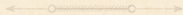
$$a^2 + b^2 = c^2$$

# Motivation

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$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



# Why Power Series Matter

- Many key functions are **defined** by power series:  $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



# Why Power Series Matter

- Many key functions are **defined** by power series:  $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$
- A power series only makes sense where it **converges**



$$f(x) = \sin x$$



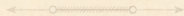
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

# Why Power Series Matter

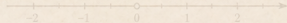
- Many key functions are **defined** by power series:  $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$
- A power series only makes sense where it **converges**
- Central question: **for which  $z$  does  $\sum c_n z^n$  converge?**



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



$$x^2 + y^2 = r^2$$

# Definition

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$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



# Definition 3.38: Power Series

## Definition 3.38

Given a sequence  $\{c_n\}$  of complex numbers, the series

$$\sum_{n=0}^{\infty} c_n z^n$$

is called a **power series**. The numbers  $c_n$  are the *coefficients*;  $z$  is a complex number.

**Key point:** convergence depends on the choice of  $z$ .



# The Circle of Convergence

## Geometric picture

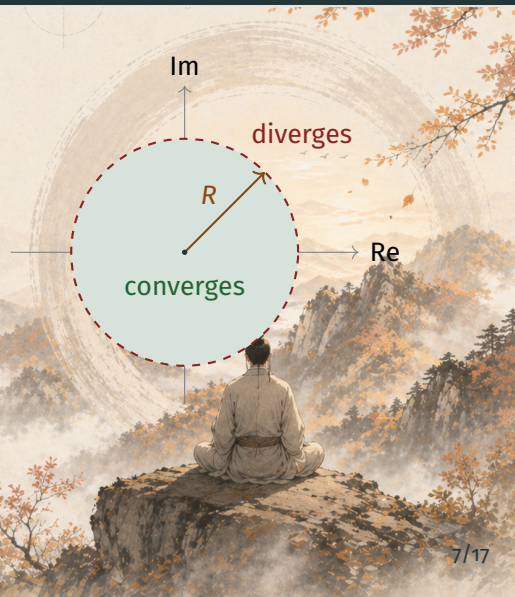
Every power series has an associated **circle of convergence** of radius  $R$ :

- converges if  $z$  is **inside**
- diverges if  $z$  is **outside**
- on the circle: *varies*



$$a^2 + b^2 = c^2$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



# The Radius of Convergence

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# Theorem 3.39: Radius of Convergence

## Theorem 3.39

Given  $\sum c_n z^n$ , put

$$\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|}, \quad R = \frac{1}{\alpha}.$$

Then  $\sum c_n z^n$  **converges** if  $|z| < R$  and **diverges** if  $|z| > R$ .

## Edge cases

If  $\alpha = 0$ , then  $R = +\infty$ ; if  $\alpha = +\infty$ , then  $R = 0$ .

# Proof of Theorem 3.39

**Strategy:** apply the **root test** (3.33) to  $a_n = c_n z^n$ .

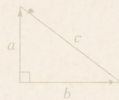
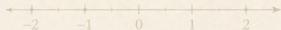
## One line

$$\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = |z| \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|} = \frac{|z|}{R}.$$

Root test: converges if  $\frac{|z|}{R} < 1$ , diverges if  $\frac{|z|}{R} > 1$ .

$$|z| < R \Rightarrow \text{converges}, \quad |z| > R \Rightarrow \text{diverges}.$$

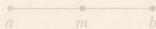
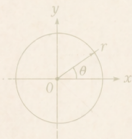




$$a^2 + b^2 = c^2$$

# Examples

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



## Example 3.40(a): $R = 0$

### Example 3.40(a)

$$\sum n^n z^n \implies R = 0.$$

Here  $\sqrt[n]{|c_n|} = \sqrt[n]{n^n} = n \rightarrow \infty$ , so  $\alpha = +\infty$  and  $R = 0$ .

**Meaning:** converges **only** at  $z = 0$ .

## Example 3.40(b): $R = +\infty$

### Example 3.40(b)

$$\sum \frac{z^n}{n!} \implies R = +\infty.$$

The ratio test is easier here:  $\frac{|z|^{n+1}/(n+1)!}{|z|^n/n!} = \frac{|z|}{n+1} \rightarrow 0$  for every  $z$ .

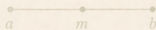
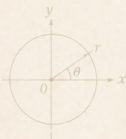
**Meaning:** converges for **all**  $z$  — this is  $e^z$ .

## Example 3.40(c): $R = 1$ , diverges on the circle

### Example 3.40(c)

$$\sum z^n \implies R = 1.$$

On the circle  $|z| = 1$ : the series **diverges**, since  $\{z^n\}$  does not tend to 0.



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



## Example 3.40(d): $R = 1$ , diverges only at $z = 1$

### Example 3.40(d)

$$\sum \frac{z^n}{n} \implies R = 1.$$

On the circle  $|z| = 1$ :

- **diverges** at  $z = 1$  (harmonic series)
- **converges** at all other  $z$  with  $|z| = 1$  (Theorem 3.44)

## Example 3.40(e): $R = 1$ , converges on the whole circle

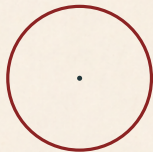
### Example 3.40(e)

$$\sum \frac{z^n}{n^2} \implies R = 1.$$

On the circle  $|z| = 1$ : **converges everywhere**, since  $\left| \frac{z^n}{n^2} \right| = \frac{1}{n^2}$ .

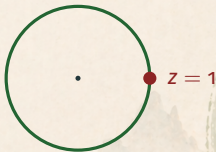
# The Boundary Circle: Three Stories ( $R = 1$ )

(c)  $\sum z^n$



diverges everywhere

(d)  $\sum z^n/n$



one bad point

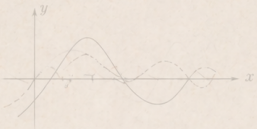
(e)  $\sum z^n/n^2$



converges everywhere

Same radius  $R = 1$  — yet **three different** boundary behaviors.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



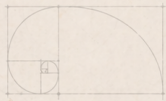
$$e^{i\theta} = \cos \theta + i \sin \theta$$



$$\nabla^2 \phi = 0$$

# Summary

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$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

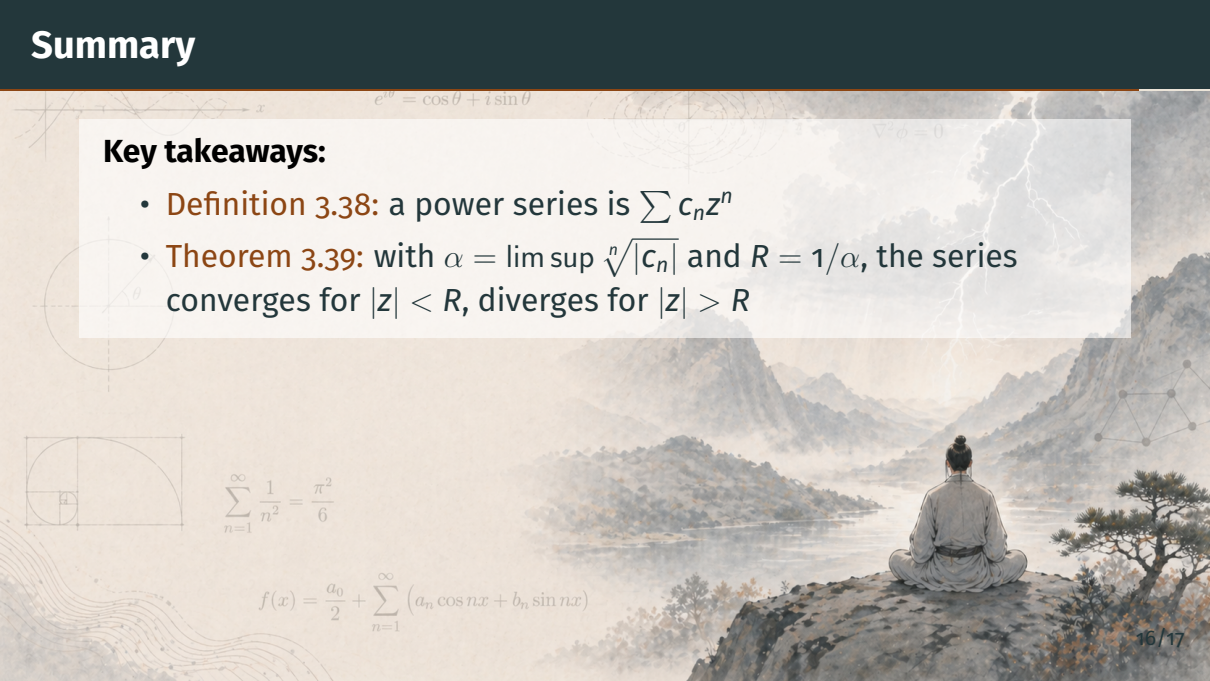
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



# Summary

## Key takeaways:

- **Definition 3.38:** a power series is  $\sum c_n z^n$
- **Theorem 3.39:** with  $\alpha = \limsup \sqrt[n]{|c_n|}$  and  $R = 1/\alpha$ , the series converges for  $|z| < R$ , diverges for  $|z| > R$

The background of the slide is a collage of various mathematical and artistic elements. At the top, there's a dark blue header with the word 'Summary' in white. Below it, a light beige banner contains the text 'Key takeaways:' followed by two bullet points. The main body of the slide features a large, misty landscape painting of a person sitting on a cliff, looking out over a river and mountains. Overlaid on this are several mathematical diagrams and equations. In the top left, there's a coordinate system with a sine wave and the equation  $e^{i\theta} = \cos \theta + i \sin \theta$ . In the top right, there's a polar coordinate system with the equation  $\nabla^2 \phi = 0$ . In the middle left, there's a diagram of a circle with a point  $\theta$  on its circumference. In the bottom left, there's a diagram of a square with a circle inside it, and the equation  $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ . In the bottom center, there's the equation  $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$ . In the bottom right, there's a diagram of a network of points connected by lines. The overall aesthetic is a blend of traditional Chinese ink wash painting and modern mathematical graphics.
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

# Summary

## Key takeaways:

- **Definition 3.38:** a power series is  $\sum c_n z^n$
- **Theorem 3.39:** with  $\alpha = \limsup \sqrt[n]{|c_n|}$  and  $R = 1/\alpha$ , the series converges for  $|z| < R$ , diverges for  $|z| > R$
- On the circle  $|z| = R$ : behavior **varies** (examples c, d, e)

**Coming up next:** Summation by Parts.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$